

Halliday & Resnick's
Fundamentals of Physics
Extended Edition **Wiley**

Chapter 15

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Oscillations

15.1 Oscillatory motion

Motion which is periodic in time, that is, motion that repeats itself in time.

Examples:

- Power line oscillates when the wind blows past it
- Earthquake oscillations move buildings

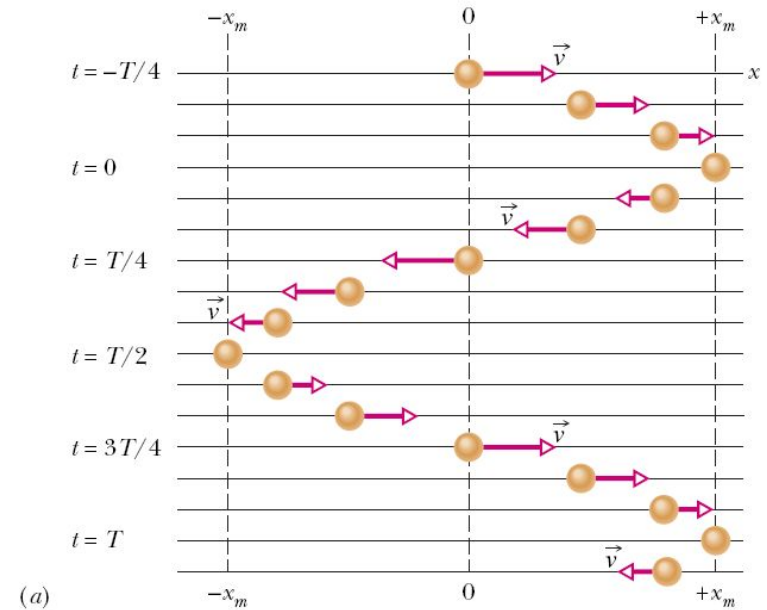
Sometimes the oscillations are so severe, that the system exhibiting oscillations break apart.

15.2 Simple Harmonic Motion

In the figure snapshots of a simple oscillatory system is shown. A particle repeatedly moves back and forth about the point $x=0$.

The time taken for one complete oscillation is the period, T . In the time of one T , the system travels from $x=+x_m$, to $-x_m$, and then back to its original position x_m .

The velocity vector arrows are scaled to indicate the magnitude of the speed of the system at different times. At $x=\pm x_m$, the velocity is zero.



15.2 Simple Harmonic Motion

Frequency of oscillation is the number of oscillations that are completed in each second.

The symbol for frequency is f , and the SI unit is the hertz (abbreviated as Hz).

It follows that

$$T = \frac{1}{f}$$

15.2 Simple Harmonic Motion

Any motion that repeats itself is periodic or harmonic.

If the motion is a sinusoidal function of time, it is called simple harmonic motion (SHM).

Mathematically SHM can be expressed as:

$$x(t) = x_m \cos(\omega t + \phi)$$

Here,

- x_m is the amplitude (maximum displacement of the system)
- t is the time
- ω is the angular frequency, and
- ϕ is the phase constant or phase angle

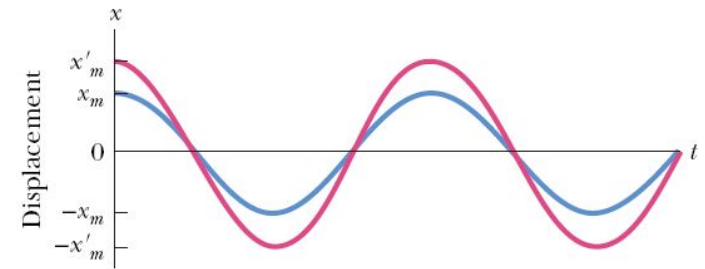
15.2 Simple Harmonic Motion

Figure a plots the displacement of two SHM systems that are different in amplitudes, but have the same period.

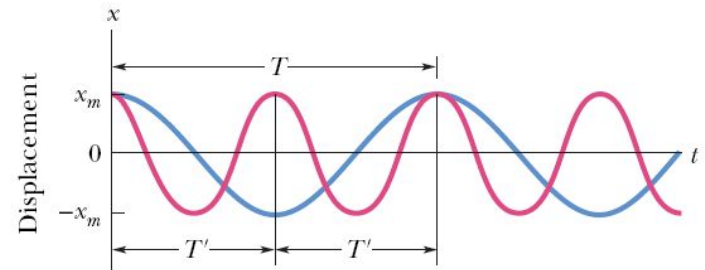
Figure b plots the displacement of two SHM systems which are different in periods but have the same amplitude.

The value of the phase constant term, ϕ , depends on the value of the displacement and the velocity of the system at time $t = 0$.

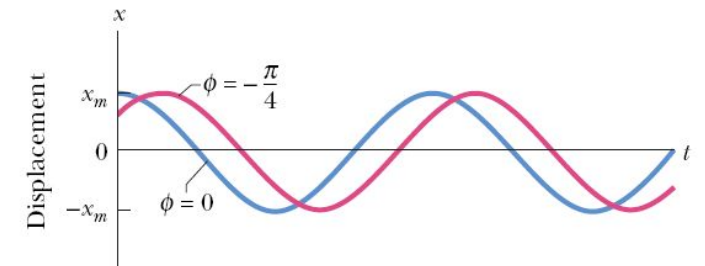
Figure c plots the displacement of two SHM systems having the same period and amplitude, but different phase constants.



(a)



(b)



(c)

15.2 Simple Harmonic Motion

For an oscillatory motion with period T ,

$$x(t) = x(t + T)$$

The cosine function also repeats itself when the argument increases by 2π . Therefore,

$$\omega(t + T) = \omega t + 2\pi$$

$$\rightarrow \omega T = 2\pi$$

$$\rightarrow \omega = \frac{2\pi}{T} = 2\pi f$$

Here, ω is the angular frequency, and measures the angle per unit time. Its SI unit is radians/second. To be consistent, ϕ then must be in radians.

15.2 Simple Harmonic Motion

The velocity of SHM:

$$v(t) = \frac{dx(t)}{dt} = \frac{d}{dt} [x_m \cos(\omega t + \phi)]$$

$$\rightarrow v(t) = -\omega x_m \sin(\omega t + \phi)$$

The maximum value (amplitude) of velocity is ωx_m . The phase shift of the velocity is $\pi/2$, making the cosine to a sine function.

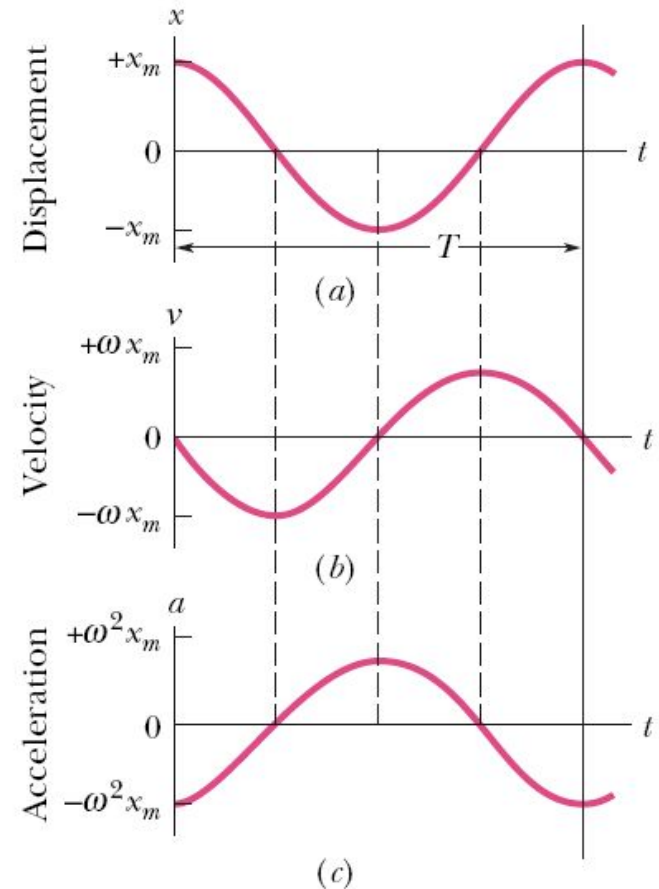
The acceleration of SHM is:

$$a(t) = \frac{dv(t)}{dt} = \frac{d}{dt} [-\omega x_m \sin(\omega t + \phi)]$$

$$\rightarrow a(t) = -\omega^2 x_m \cos(\omega t + \phi)$$

$$\rightarrow a(t) = -\omega^2 x(t)$$

The acceleration amplitude is $\omega^2 x_m$.



In SHM $a(t)$ is proportional to the displacement but opposite in sign.

15.3 Force Law for SHM

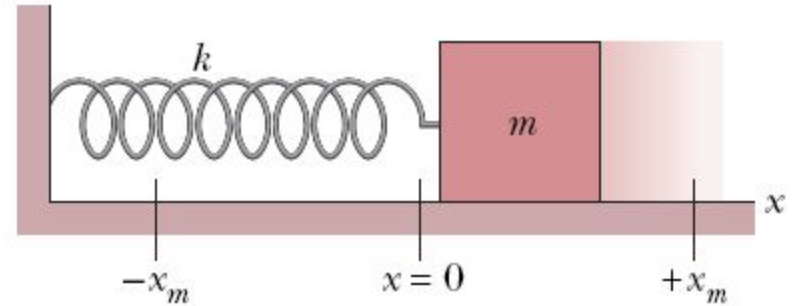
From Newton's 2nd law:

$$F = ma = -(m\omega^2)x = -kx$$

SHM is the motion executed by a system subject to a force that is proportional to the displacement of the system but opposite in sign.

15.3 Force Law for SHM

The block-spring system shown on the right forms a linear SHM oscillator.



The spring constant of the spring, k , is related to the angular frequency, ω , of the oscillator:

$$\omega = \sqrt{\frac{k}{m}} \rightarrow T = 2\pi \sqrt{\frac{m}{k}}$$

Example, force law: A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(a) What are the angular frequency, the frequency, and the period of the resulting motion?

KEY IDEA

The block–spring system forms a linear simple harmonic oscillator, with the block undergoing SHM.

Calculations: The angular frequency is given by Eq. 15-12:

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{65 \text{ N/m}}{0.68 \text{ kg}}} = 9.78 \text{ rad/s} \\ \approx 9.8 \text{ rad/s.} \quad (\text{Answer})$$

The frequency follows from Eq. 15-5, which yields

$$f = \frac{\omega}{2\pi} = \frac{9.78 \text{ rad/s}}{2\pi \text{ rad}} = 1.56 \text{ Hz} \approx 1.6 \text{ Hz.} \quad (\text{Answer})$$

The period follows from Eq. 15-2, which yields

$$T = \frac{1}{f} = \frac{1}{1.56 \text{ Hz}} = 0.64 \text{ s} = 640 \text{ ms.} \quad (\text{Answer})$$

Example, force law:

A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(b) What is the amplitude of the oscillation?

KEY IDEA

With no friction involved, the mechanical energy of the spring–block system is conserved.

Reasoning: The block is released from rest 11 cm from its equilibrium position, with zero kinetic energy and the elastic potential energy of the system at a maximum. Thus, the block will have zero kinetic energy whenever it is again 11 cm from its equilibrium position, which means it will never be farther than 11 cm from that position. Its maximum displacement is 11 cm:

$$x_m = 11 \text{ cm.} \quad (\text{Answer})$$

Example, force law: A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(c) What is the maximum speed v_m of the oscillating block, and where is the block when it has this speed?

KEY IDEA The maximum speed v_m is the velocity amplitude ωx_m in Eq. 15-6.

Calculation: Thus, we have

$$\begin{aligned} v_m &= \omega x_m = (9.78 \text{ rad/s})(0.11 \text{ m}) \\ &= 1.1 \text{ m/s.} \end{aligned} \quad (\text{Answer})$$

This maximum speed occurs when the oscillating block is rushing through the origin; compare Figs. 15-4*a* and 15-4*b*, where you can see that the speed is a maximum whenever $x = 0$.

Example, force law:

A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(d) What is the magnitude a_m of the maximum acceleration of the block?

KEY IDEA

The magnitude a_m of the maximum acceleration is the acceleration amplitude $\omega^2 x_m$ in Eq. 15-7.

Calculation: So, we have

$$\begin{aligned} a_m &= \omega^2 x_m = (9.78 \text{ rad/s})^2 (0.11 \text{ m}) \\ &= 11 \text{ m/s}^2. \end{aligned} \quad (\text{Answer})$$

This maximum acceleration occurs when the block is at the ends of its path. At those points, the force acting on the block has its maximum magnitude; compare Figs. 15-4*a* and 15-4*c*, where you can see that the magnitudes of the displacement and acceleration are maximum at the same times.

Example, force law:

A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(e) What is the phase constant ϕ for the motion?

Calculations: Equation 15-3 gives the displacement of the block as a function of time. We know that at time $t = 0$, the block is located at $x = x_m$. Substituting these *initial conditions*, as they are called, into Eq. 15-3 and canceling x_m give us

$$1 = \cos \phi. \quad (15-14)$$

Taking the inverse cosine then yields

$$\phi = 0 \text{ rad.} \quad (\text{Answer})$$

(Any angle that is an integer multiple of 2π rad also satisfies Eq. 15-14; we chose the smallest angle.)

Example, force law:

A block whose mass m is 680 g is fastened to a spring whose spring constant k is 65 N/m. The block is pulled a distance $x = 11$ cm from its equilibrium position at $x = 0$ on a frictionless surface and released from rest at $t = 0$.

(f) What is the displacement function $x(t)$ for the spring–block system?

Calculation: The function $x(t)$ is given in general form by Eq. 15-3. Substituting known quantities into that equation gives us

$$\begin{aligned}x(t) &= x_m \cos(\omega t + \phi) \\&= (0.11 \text{ m}) \cos[(9.8 \text{ rad/s})t + 0] \\&= 0.11 \cos(9.8t), \quad (\text{Answer})\end{aligned}$$

where x is in meters and t is in seconds.

15.4: An Angular SHM

The figure shows an example of angular SHM. In a torsion pendulum involves the twisting of a suspension wire as the disk oscillates in a horizontal plane.

The torque associated with an angular displacement of θ is given by:

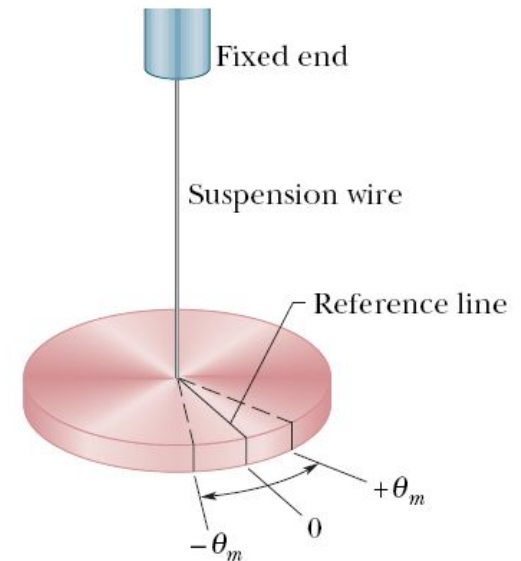
$$\tau = -\kappa\theta$$

κ is the torsion constant, and depends on the length, diameter, and material of the suspension wire.

The period, T , relates to κ as:

$$T = 2\pi\sqrt{\frac{I}{\kappa}}$$

Here, I is the rotational inertia of the oscillating disk.



15.6: Pendulums

In a *simple pendulum*, a particle of mass m is suspended from one end of an unstretchable massless string of length L that is fixed at the other end.

The restoring torque acting on the mass when its angular displacement is θ , is:

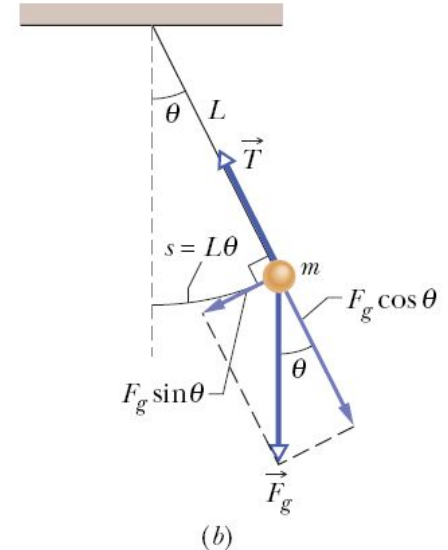
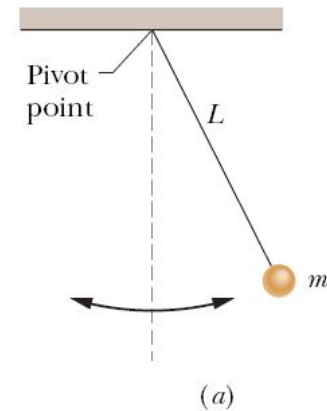
$$\tau = -L(F_g \sin \theta) = I\alpha$$

α is the angular acceleration of the mass. Finally,

$$\alpha = -\frac{mgL}{I}\theta, \text{ and}$$

$$T = 2\pi\sqrt{\frac{L}{g}}$$

This is true for *small angular displacements*, θ .



15.7: SHM and uniform circular motion

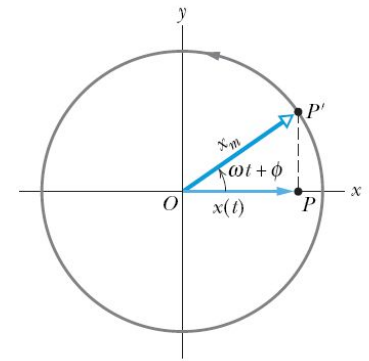
Consider a reference particle P' moving in uniform circular motion with constant angular speed (ω).

The projection of the particle on the x-axis is a point P , describing motion given by:

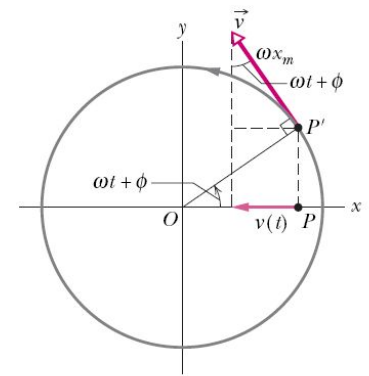
$$x(t) = x_m \cos(\omega t + \phi).$$

This is the displacement equation of SHM.

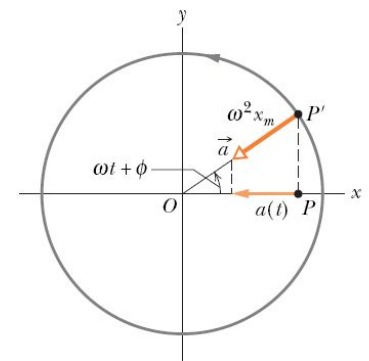
SHM, therefore, is the projection of uniform circular motion on a diameter of the circle in which the circular motion occurs.



(a)



(b)



(c)